

Confinement enhanced clustering in dense active systems

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We study the effect of confinement in dense active lattice gases in one and two dimensions. In the one dimensional case we study hard-core active particles on a lattice which can be in two orientations, positive or negative. We derive the probability distribution for this active particle exactly. When a finite density of such particles are introduced in the system, we find that starting from a step initial condition, the normalized probability of finding a particle at the origin is a non-monotonic function of the number of walkers. The peak of the probability as a function of the number of walkers is biased outwards with increasing the time, leading to an effective outward pressure. In the case of the two dimensional system, we study a single active particle on the square as well as triangular lattice, where the particles can take on four and six orientations respectively. We derive the distribution of the position of the walker exactly. Finally we study the effect of confinement in the two dimensional model, and show that the activity enhanced clustering leads to non-monotonic behaviour of the probability of particles at the origin as a function of the density.

I. INTRODUCTION

Active matter, consisting of particles that convert ambient energy to directed motion, is an important class of non-equilibrium systems that describe several naturally relevant systems. Examples range from synthetic colloids to bacterial suspensions. Fluctuation-dissipation relations do not generally apply to such systems since the driving in these systems exist at microscopic scales than thermal systems. A particular collective behavior that has been widely studied is motility-induced phase separation (MIPS). MIPS, a kinetic phenomenon, is striking in its similarity to equilibrium phase separation such as in passive colloids with attractive interactions. The non-equilibrium phases and transitions between them while having exact analogs in equilibrium systems. However questions regarding the universality of MIPS as a general mechanism of non equilibrium phase separation still remain. In particular the effect of the microscopically driven dynamics on the effective attraction that leads to clustering in such systems still remains. In this context it is important to study the behaviour of simple models that can lead to a deeper understanding of the clustering phenomena in such systems.

In this paper we study the effect of confinement in dense active lattice gases in one and two dimensions. In the one dimensional case we study hard-core active particles on a lattice which can be in two orientations, positive or negative. We derive the probability distribution for this active particle exactly. When a finite density of such particles are introduced in the system, we find that starting from a step initial condition, the normalized probability of finding a particle at the origin is a non-monotonic function of the number of walkers. The peak of the probability as a function of the number of

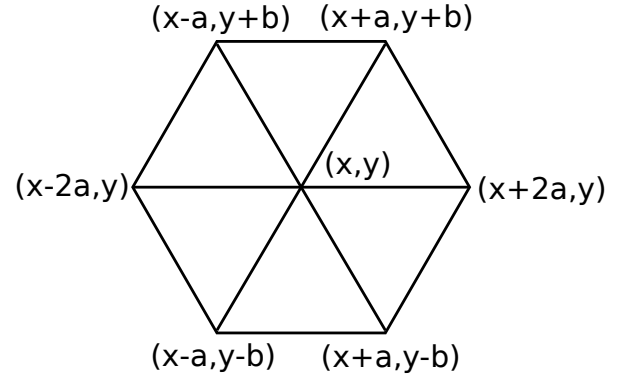


FIG. 1. The various lattices used in our study. [Make a better schematic figure of 1d, 2d, square, triangular lattices.](#)

walkers is biased outwards with increasing the time, leading to an effective outward pressure. In the case of the two dimensional system, we study a single active particle on the square as well as triangular lattice, where the particles can take on four and six orientations respectively. We derive the distribution of the position of the walker exactly. Finally we study the effect of confinement in the two dimensional model, and show that the activity enhanced clustering leads to non-monotonic behaviour of the probability of particles at the origin as a function of the density.

II. SINGLE WALKER ON A 1D LATTICE

Consider a random walker on one dimensional lattice of length L . The walker may have one of two possible states, which are positive and negative state. Let $P_+(x, t)$ and $P_-(x, t)$ respectively represent the probability of finding the walker at position x at time t . The translational rate for the particle with positive state along positive and negative x directions are respectively $(\frac{1}{2} + \epsilon)$ and $(\frac{1}{2} - \epsilon)$. The translational rates for the particle with negative state are

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vice versa of the rates of the particle with positive state. Along with the translation, the particle may flip its state at the rate γ . The equations obeyed by $P_{\pm}(x, t)$ may be written as

$$\frac{\partial P_+(x, t)}{\partial t} = \left(\frac{1}{2} + \epsilon\right)P_+(x-1, t) + \left(\frac{1}{2} - \epsilon\right)P_+(x+1, t) - P_+(x, t) + \gamma P_-(x, t) - \gamma P_+(x, t), \quad (1)$$

$$\frac{\partial P_-(x, t)}{\partial t} = \left(\frac{1}{2} + \epsilon\right)P_-(x+1, t) + \left(\frac{1}{2} - \epsilon\right)P_-(x-1, t) - P_-(x, t) + \gamma P_+(x, t) - \gamma P_-(x, t). \quad (2)$$

Fourier transform of $P_{\pm}(x, t)$ may be defined as

$$\tilde{P}_{\pm}(n, t) = \sum_{x=0}^{L-1} e^{i\frac{2\pi}{L}nx} P_{\pm}(x, t). \quad (3)$$

Applying the Fourier transform Eq. (A1) and Eq. (A2) may be written as

$$\frac{\partial \tilde{P}_+(k, t)}{\partial t} = [\cos(k) - 1 - \gamma + i2\epsilon \sin(k)]\tilde{P}_+(k, t) + \gamma \tilde{P}_-(k, t), \quad (4)$$

$$\frac{\partial \tilde{P}_-(k, t)}{\partial t} = [\cos(k) - 1 - \gamma - i2\epsilon \sin(k)]\tilde{P}_-(k, t) + \gamma \tilde{P}_+(k, t), \quad (5)$$

where $k = \frac{2\pi n}{L}$. In matrix form we may write

$$\frac{\partial}{\partial t} \begin{pmatrix} \tilde{P}_+(k, t) \\ \tilde{P}_-(k, t) \end{pmatrix} = \mathcal{M}(k) \begin{pmatrix} \tilde{P}_+(k, t) \\ \tilde{P}_-(k, t) \end{pmatrix}, \quad (6)$$

where

$$\mathcal{M}(k) = \begin{pmatrix} a(k) & \gamma \\ \gamma & a^*(k) \end{pmatrix}, \quad (7)$$

and the coefficient is $a(k) = \cos(k) - 1 - \gamma + i2\epsilon \sin(k)$. Let λ_1 and λ_2 are two eigen values of the matrix $\mathcal{M}(k)$. The eigen values may be expressed as

$$\lambda_{1,2} = \cos(k) - 1 - \gamma \pm \sqrt{\gamma^2 - 4\epsilon^2 \sin^2(k)}. \quad (8)$$

Solution of Eq. (A6) may be written as

$$\vec{P}(k, t) = c_1 \vec{X}_1 e^{\lambda_1 t} + c_2 \vec{X}_2 e^{\lambda_2 t}, \quad (9)$$

where $\vec{X}_1$ and $\vec{X}_2$ are eigen vectors of matrix $\mathcal{M}$ corresponding to the eigen values λ_1 and λ_2 respectively, and c_1 and c_2 are the prefactors to be determined from the initial state. We take the initial condition $P_+(x, t=0) = \frac{1}{2}\delta_{x,x'}$ and $P_-(x, t=0) = \frac{1}{2}\delta_{x,x'}$. Total probability may be written as

$$\begin{aligned} \tilde{P}(n, t) &= \tilde{P}_+(n, t) + \tilde{P}_-(n, t) \\ &= a_1 e^{\lambda_1 t} + a_2 e^{\lambda_2 t}, \end{aligned} \quad (10)$$

where

$$\begin{aligned} a_1 &= \frac{\gamma + \sqrt{\gamma^2 - 4\epsilon^2 \sin^2(k)}}{2\sqrt{\gamma^2 - 4\epsilon^2 \sin^2(k)}}, \\ a_2 &= \frac{\sqrt{\gamma^2 - 4\epsilon^2 \sin^2(k)} - \gamma}{2\sqrt{\gamma^2 - 4\epsilon^2 \sin^2(k)}}. \end{aligned}$$

To get the probability distribution in position space we have to perform the inverse Fourier transformation

$$P(x, t) = \sum_{n=0}^{L-1} e^{i\frac{2\pi}{L}nx} \tilde{P}(n, t). \quad (11)$$

Can we get a closed form for this inverse Fourier transform. This should match the Malakar result in the continuum limit.

Look at the saddle point result in the Urna/Sanjib paper.

III. MULTIPLE WALKERS ON A 1D LATTICE

A. Multiple Walkers with step initial conditions

Consider W walkers sitting on lattice with step initial condition i.e. at time $t = 0$, the particles are confined symmetrically about the point $x = L/2$. The normalized probability distribution of the system at time $t = 0$ is a step function of width W and height $1/W$. The normalized probability of finding a particle at the centre i.e. $x = L/2$ for unbiased walkers ($\epsilon = 0$) decreases monotonically with increasing particle density as shown in fig. 2(a). Now, we consider active biased random walkers i.e. $\gamma \neq 0$ and $\epsilon \neq 0$. The normalized probability distribution of the system for $\gamma = 0.05$ and $\gamma = 0.005$ with $\epsilon = 0.2$ at $t = 60$ are shown in fig. 2(b) and (c) respectively. We see that unlike the unbiased random walkers the probability of finding a particle at the origin for active biased random walkers show non-monotonic behaviour. We have shown the probability density at the centre as a function of number of walkers W in fig. 2(d) for different times t . We see that the peak of the curve for fixed time shifts towards the right side with increasing the time.

B. Multiple Walkers with random initial conditions

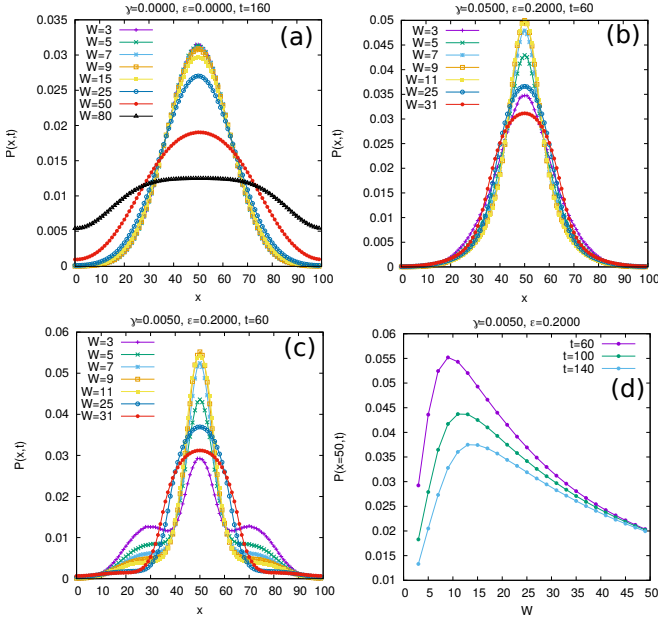


FIG. 2. Probability distribution of (a) unbiased random walker ($\epsilon = 0$) at fixed time $t = 160$, active biased random walker (b) ($\gamma = 0.05$ and $\epsilon = 0.2$) (c) ($\gamma = 0.005$ and $\epsilon = 0.2$) at fixed time $t = 60$ for different particle densities. (d) Probability density at centre vs the number of walker W for different times. Can we try a scaling collapse of (d)?

Have we tried this?

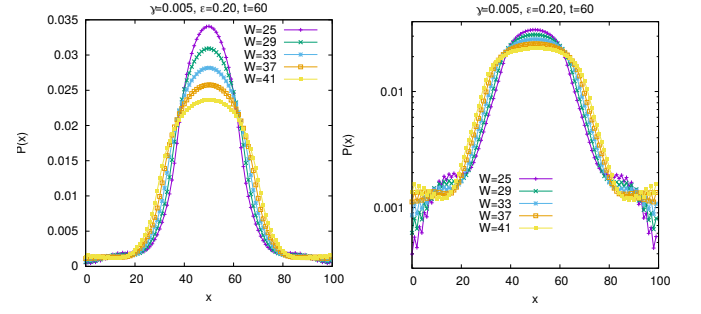


FIG. 3. Multiple walker on one dimensional lattice with discrete time where W is the number of random walkers.

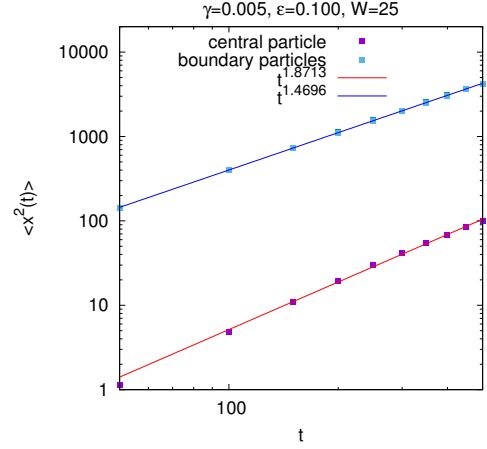


FIG. 4. Plot of mean square displacement vs. time of the random walkers sitting at the central position and two boundary positions. This behaviour is surprising, but probably due to the clustering near the extremes. How does this change if activity is lowered. Do we know the answer for the passive case?

C. Coarsening Properties

The density-density correlation $C(r, t)$ may be defined as

$$C(r, t) = \frac{1}{L} \sum_{i=0}^{L-1} \eta(i) \eta(i+r), \quad (12)$$

where $\eta(i)$ is the density field at i -th lattice position. The field $\eta(i)$ takes nonzero unit values if the position is occupied by a particle, otherwise it is zero. The plot of correlation function for various particle densities ρ are shown in fig. 5(a) and (b) for system sizes $L = 100$ and $L = 500$ respectively.

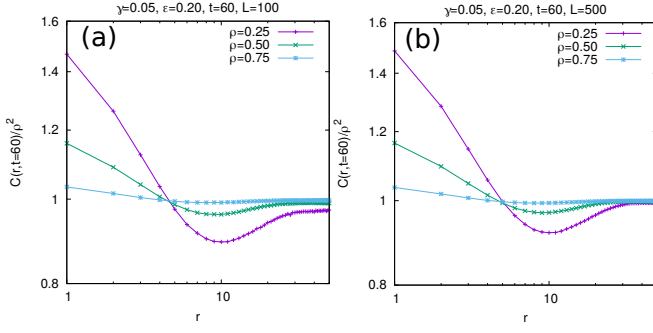


FIG. 5. Scaled density-density correlation plot with varying particle density ρ for system size (a) $L=100$ and (b) $L=500$.

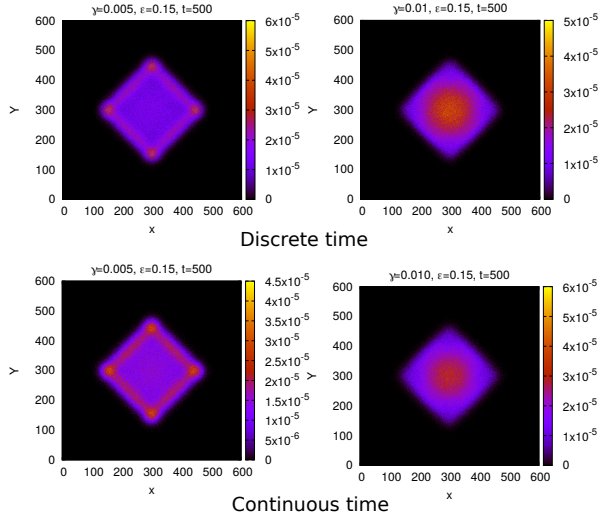


FIG. 6. Comparison of discrete time and continuous time random walk on square lattice.

IV. SINGLE WALKER ON A 2D LATTICE

A. Square Lattice

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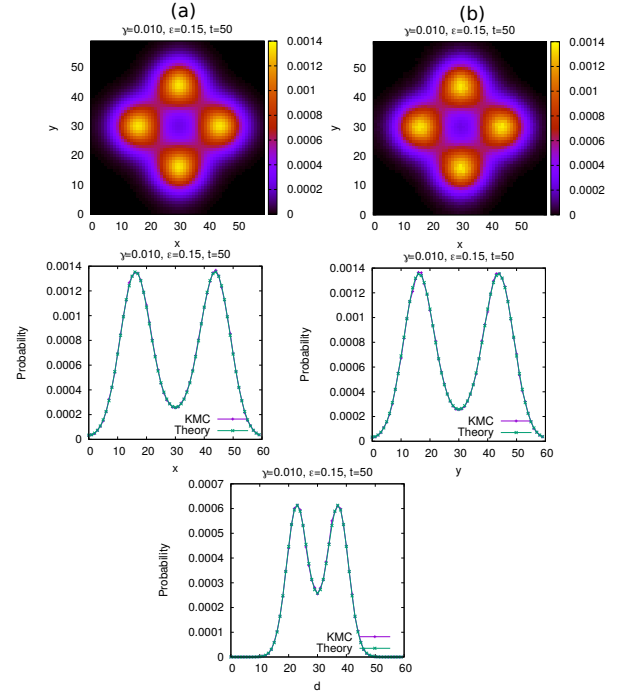


FIG. 7. Comparison of (a) theory and (b) kinetic Monte Carlo result of active random walk on square lattice.

B. Triangular Lattice

Maybe write the shortened versions from the appendix here.

V. MULTIPLE WALKERS ON A 2D LATTICE

We next analyze the case of multiple particles on two dimensional lattices. We focus on the square lattice case, with different initial conditions and study the coarsening properties as for the one dimensional case.

A. Multiple Walkers with step initial conditions

B. Multiple Walkers with random initial conditions

C. Coarsening Properties

VI. DISCUSSION

We have studied the probability distribution of multiple random walker with activity and bias on one dimensional lattice. The normalized probability of finding a particle at the centre at a fixed time shows non-monotonic behavior as we increase the number of walkers. We have also shown that the peak of the curve shifts towards right side with increasing time.

Appendix A: One dimensional lattice

Consider a random walker on one dimensional lattice of length L . The walker may have one of two possible states positive-state and negative-state. Let, $P_+(x, t)$ and $P_-(x, t)$ respectively represent the probability of finding the walker at position x at time t . The equations obeyed by $P_\pm(x, t)$ are

$$\frac{\partial P_+(x, t)}{\partial t} = \left(\frac{1}{2} + \epsilon\right)P_+(x-1, t) + \left(\frac{1}{2} - \epsilon\right)P_+(x+1, t) - P_+(x, t) + \gamma P_-(x, t) - \gamma P_+(x, t), \quad (\text{A1})$$

$$\frac{\partial P_-(x, t)}{\partial t} = \left(\frac{1}{2} + \epsilon\right)P_-(x+1, t) + \left(\frac{1}{2} - \epsilon\right)P_-(x-1, t) - P_-(x, t) + \gamma P_+(x, t) - \gamma P_-(x, t). \quad (\text{A2})$$

Fourier transform is defined as

$$\tilde{P}_\pm(n, t) = \sum_{x=0}^{L-1} e^{i\frac{2\pi}{L}nx} P_\pm(x, t). \quad (\text{A3})$$

Applying Fourier transform, Eq. A1 and Eq. A2 may be written as

$$\frac{\partial \tilde{P}_+(k, t)}{\partial t} = [\cos(k) - 1 - \gamma + i2\epsilon \sin(k)]\tilde{P}_+(k, t) + \gamma \tilde{P}_-(k, t), \quad (\text{A4})$$

$$\frac{\partial \tilde{P}_-(k, t)}{\partial t} = \gamma \tilde{P}_+(k, t) + [\cos(k) - 1 - \gamma - i2\epsilon \sin(k)]\tilde{P}_-(k, t), \quad (\text{A5})$$

$k = \frac{2\pi n}{L}$. In matrix form we can write

$$\frac{\partial}{\partial t} \begin{pmatrix} \tilde{P}_+(k, t) \\ \tilde{P}_-(k, t) \end{pmatrix} = \begin{pmatrix} \cos(k) - 1 - \gamma + i2\epsilon \sin(k) & \gamma \\ \gamma & \cos(k) - 1 - \gamma - i2\epsilon \sin(k) \end{pmatrix} \begin{pmatrix} \tilde{P}_+(k, t) \\ \tilde{P}_-(k, t) \end{pmatrix}. \quad (\text{A6})$$

Let λ_1 and λ_2 are two eigen values of the matrix

$$M = \begin{pmatrix} \cos(k) - 1 - \gamma + i2\epsilon \sin(k) & \gamma \\ \gamma & \cos(k) - 1 - \gamma - i2\epsilon \sin(k) \end{pmatrix}. \quad (\text{A7})$$

The eigenvalues may be written as

$$\lambda_{1,2} = \cos(k) - 1 - \gamma \pm \sqrt{\gamma^2 - 4\epsilon^2 \sin^2(k)}. \quad (\text{A8})$$

Solution of Eq. A6 may be written as

$$\vec{P}(k, t) = c_1 \vec{X}_1 e^{\lambda_1 t} + c_2 \vec{X}_2 e^{\lambda_2 t}, \quad (\text{A9})$$

where $\vec{X}_1$ and $\vec{X}_2$ are eigen vectors of matrix M corresponding to the eigen values λ_1 and λ_2 respectively, and c_1 and c_2 are the prefactors to be determined from the initial state. We take the initial condition $P_+(x, t=0) = \frac{1}{2}\delta_{x,x'}$ and $P_-(x, t=0) = \frac{1}{2}\delta_{x,x'}$. Total probability may be written as

$$\begin{aligned} \tilde{P}(n, t) &= \tilde{P}_+(n, t) + \tilde{P}_-(n, t) \\ &= \frac{\gamma + \sqrt{\gamma^2 - 4\epsilon^2 \sin^2(k)}}{2\sqrt{\gamma^2 - 4\epsilon^2 \sin^2(k)}} e^{\lambda_1 t} + \frac{\sqrt{\gamma^2 - 4\epsilon^2 \sin^2(k)} - \gamma}{2\sqrt{\gamma^2 - 4\epsilon^2 \sin^2(k)}} e^{\lambda_2 t} \end{aligned} \quad (\text{A10})$$

To get the probability distribution in position space we have to do inverse transformation

$$P(x, t) = \sum_{n=0}^{L-1} e^{i\frac{2\pi}{L}nx} \tilde{P}(n, t) \quad (\text{A11})$$

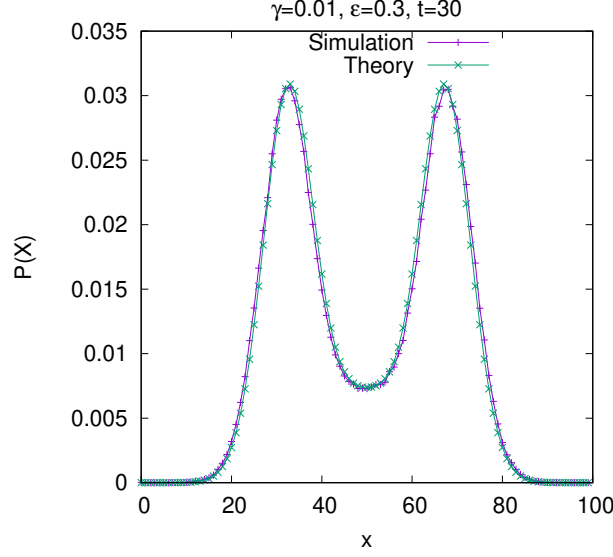


FIG. 8. Comparison of continuous time kinetic Monte Carlo simulation results with the theory results obtained from Eq. A11.

1. Two dimensional square lattice

Consider a random walker on two dimensional square lattice. The walker can have one of the four possible states 0, 1, 2 and 3. The walk is biased along positive x , positive y , negative x or negative y direction if the walker is 0, 1, 2 or 3 state respectively. The probabilities of four states obey the equations

$$\begin{aligned} \frac{\partial P_0(x, y, t)}{\partial t} = & \left(\frac{1}{4} + \epsilon\right)P_0(x-1, y, t) + \left(\frac{1}{4} - \epsilon\right)P_0(x+1, y, t) + \frac{1}{4}P_0(x, y-1, t) + \frac{1}{4}P_0(x, y+1, t) - P_0(x, y, t) \\ & + \frac{\gamma}{2}P_1(x, y, t) + \frac{\gamma}{2}P_3(x, y, t) - \gamma P_0(x, y, t), \end{aligned} \quad (\text{A12})$$

$$\begin{aligned} \frac{\partial P_1(x, y, t)}{\partial t} = & \frac{1}{4}P_1(x-1, y, t) + \frac{1}{4}P_1(x+1, y, t) + \left(\frac{1}{4} + \epsilon\right)P_1(x, y-1, t) + \left(\frac{1}{4} - \epsilon\right)P_1(x, y+1, t) - P_1(x, y, t) \\ & + \frac{\gamma}{2}P_2(x, y, t) + \frac{\gamma}{2}P_0(x, y, t) - \gamma P_1(x, y, t), \end{aligned} \quad (\text{A13})$$

$$\begin{aligned} \frac{\partial P_2(x, y, t)}{\partial t} = & \left(\frac{1}{4} - \epsilon\right)P_2(x-1, y, t) + \left(\frac{1}{4} + \epsilon\right)P_2(x+1, y, t) + \frac{1}{4}P_2(x, y-1, t) + \frac{1}{4}P_2(x, y+1, t) - P_2(x, y, t) \\ & + \frac{\gamma}{2}P_3(x, y, t) + \frac{\gamma}{2}P_1(x, y, t) - \gamma P_2(x, y, t), \end{aligned} \quad (\text{A14})$$

$$\begin{aligned} \frac{\partial P_3(x, y, t)}{\partial t} = & \frac{1}{4}P_3(x-1, y, t) + \frac{1}{4}P_3(x+1, y, t) + \left(\frac{1}{4} - \epsilon\right)P_3(x, y-1, t) + \left(\frac{1}{4} + \epsilon\right)P_3(x, y+1, t) - P_3(x, y, t) \\ & + \frac{\gamma}{2}P_0(x, y, t) + \frac{\gamma}{2}P_2(x, y, t) - \gamma P_3(x, y, t). \end{aligned} \quad (\text{A15})$$

We define the Fourier transform of the probability distribution in i -th state on a two dimensional square lattice as

$$\tilde{P}_i(n_1, n_2, t) = \sum_{x=0}^{L-1} \sum_{y=0}^{L-1} e^{i\frac{2\pi}{L}(n_1x+n_2y)} P_i(x, y, t). \quad (\text{A16})$$

Using Fourier transform, Eqs. A12-A15 may be written as

$$\frac{\partial}{\partial t} \begin{pmatrix} \tilde{P}_0(k_1, k_2, t) \\ \tilde{P}_1(k_1, k_2, t) \\ \tilde{P}_2(k_1, k_2, t) \\ \tilde{P}_3(k_1, k_2, t) \end{pmatrix} = \begin{pmatrix} a & \gamma/2 & 0 & \gamma/2 \\ \gamma/2 & b & \gamma/2 & 0 \\ 0 & \gamma/2 & a^* & \gamma/2 \\ \gamma/2 & 0 & \gamma/2 & b^* \end{pmatrix} \begin{pmatrix} \tilde{P}_0(k_1, k_2, t) \\ \tilde{P}_1(k_1, k_2, t) \\ \tilde{P}_2(k_1, k_2, t) \\ \tilde{P}_3(k_1, k_2, t) \end{pmatrix}, \quad (\text{A17})$$

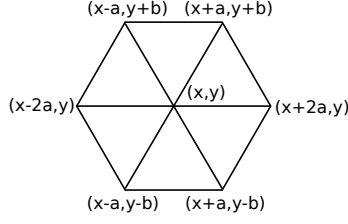


FIG. 9. Schematic diagram of the triangular lattice.

where

$$a = \frac{1}{2}[\cos(k_1) + \cos(k_2)] - 1 - \gamma + i2\epsilon \sin(k_1), \quad (\text{A18})$$

$$b = \frac{1}{2}[\cos(k_1) + \cos(k_2)] - 1 - \gamma + i2\epsilon \sin(k_2). \quad (\text{A19})$$

Appendix B: Triangular lattice

We define the Fourier transform of the probability distribution in i -th state on a two dimensional triangular lattice as

$$\tilde{P}_i(n_1, n_2, t) = \sum_x \sum_y e^{i(\frac{\pi n_1 x}{aL} + \frac{2\pi n_2 y}{bL})} P_i(x, y, t). \quad (\text{B1})$$

$$\begin{aligned} \frac{\partial P_0(x, y, t)}{\partial t} = & \left(\frac{1}{6} - \epsilon\right)P_0(x+2a, y, t) + \frac{1}{6}P_0(x+a, y+b, t) + \frac{1}{6}P_0(x-a, y+b, t) + \left(\frac{1}{6} + \epsilon\right)P_0(x-2a, y, t) \\ & + \frac{1}{6}P_0(x-a, y-b, t) + \frac{1}{6}P_0(x+a, y-b, t) - P_0(x, y, t) + \frac{\gamma}{2}P_1(x, y, t) + \frac{\gamma}{2}P_5(x, y, t) \\ & - \gamma P_0(x, y, t), \end{aligned} \quad (\text{B2})$$

$$\begin{aligned} \frac{\partial P_1(x, y, t)}{\partial t} = & \frac{1}{6}P_1(x+2a, y, t) + \left(\frac{1}{6} - \epsilon\right)P_1(x+a, y+b, t) + \frac{1}{6}P_1(x-a, y+b, t) + \frac{1}{6}P_1(x-2a, y, t) \\ & + \left(\frac{1}{6} + \epsilon\right)P_1(x-a, y-b, t) + \frac{1}{6}P_1(x+a, y-b, t) - P_1(x, y, t) + \frac{\gamma}{2}P_2(x, y, t) + \frac{\gamma}{2}P_0(x, y, t) \\ & - \gamma P_1(x, y, t), \end{aligned} \quad (\text{B3})$$

$$\begin{aligned} \frac{\partial P_2(x, y, t)}{\partial t} = & \frac{1}{6}P_2(x+2a, y, t) + \frac{1}{6}P_2(x+a, y+b, t) + \left(\frac{1}{6} - \epsilon\right)P_2(x-a, y+b, t) + \frac{1}{6}P_2(x-2a, y, t) \\ & + \frac{1}{6}P_2(x-a, y-b, t) + \left(\frac{1}{6} + \epsilon\right)P_2(x+a, y-b, t) - P_2(x, y, t) + \frac{\gamma}{2}P_3(x, y, t) + \frac{\gamma}{2}P_1(x, y, t) \\ & - \gamma P_2(x, y, t), \end{aligned} \quad (\text{B4})$$

$$\begin{aligned} \frac{\partial P_3(x, y, t)}{\partial t} = & \left(\frac{1}{6} + \epsilon\right)P_3(x+2a, y, t) + \frac{1}{6}P_3(x+a, y+b, t) + \frac{1}{6}P_3(x-a, y+b, t) + \left(\frac{1}{6} - \epsilon\right)P_3(x-2a, y, t) \\ & + \frac{1}{6}P_3(x-a, y-b, t) + \frac{1}{6}P_3(x+a, y-b, t) - P_3(x, y, t) + \frac{\gamma}{2}P_4(x, y, t) + \frac{\gamma}{2}P_2(x, y, t) \\ & - \gamma P_3(x, y, t), \end{aligned} \quad (\text{B5})$$

$$\begin{aligned} \frac{\partial P_4(x, y, t)}{\partial t} = & \frac{1}{6}P_4(x+2a, y, t) + \left(\frac{1}{6} + \epsilon\right)P_4(x+a, y+b, t) + \frac{1}{6}P_4(x-a, y+b, t) + \frac{1}{6}P_4(x-2a, y, t) \\ & + \left(\frac{1}{6} - \epsilon\right)P_4(x-a, y-b, t) + \frac{1}{6}P_4(x+a, y-b, t) - P_4(x, y, t) + \frac{\gamma}{2}P_5(x, y, t) + \frac{\gamma}{2}P_3(x, y, t) \\ & - \gamma P_4(x, y, t), \end{aligned} \quad (\text{B6})$$

$$\begin{aligned} \frac{\partial P_5(x, y, t)}{\partial t} = & \frac{1}{6}P_5(x+2a, y, t) + \frac{1}{6}P_5(x+a, y+b, t) + \left(\frac{1}{6} + \epsilon\right)P_5(x-a, y+b, t) + \frac{1}{6}P_5(x-2a, y, t) \\ & + \frac{1}{6}P_5(x-a, y-b, t) + \left(\frac{1}{6} - \epsilon\right)P_5(x+a, y-b, t) - P_5(x, y, t) + \frac{\gamma}{2}P_0(x, y, t) + \frac{\gamma}{2}P_4(x, y, t) \\ & - \gamma P_5(x, y, t). \end{aligned} \quad (\text{B7})$$

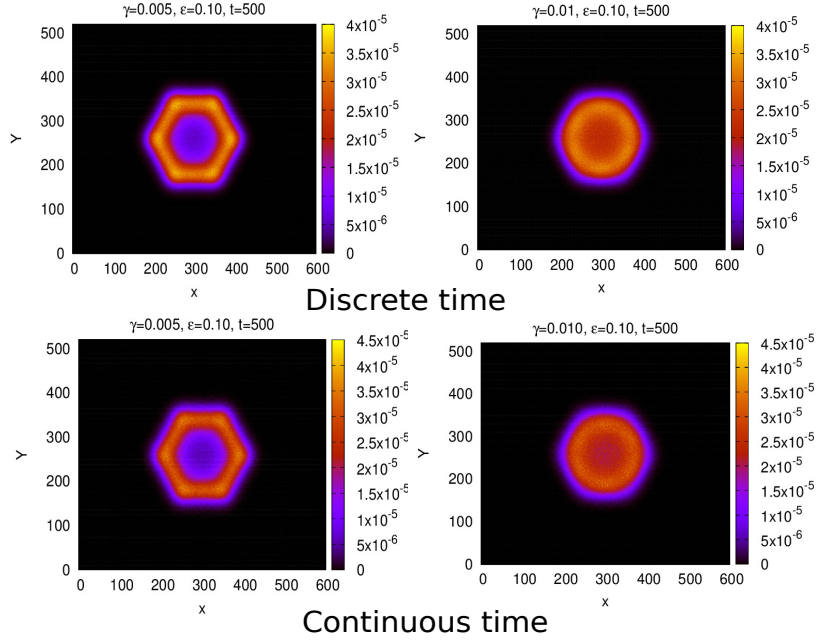


FIG. 10. Comparison of discrete time and continuous time random walk on triangular lattice.

$$\frac{\partial}{\partial t} \begin{pmatrix} \tilde{P}_0(k_1, k_2, t) \\ \tilde{P}_1(k_1, k_2, t) \\ \tilde{P}_2(k_1, k_2, t) \\ \tilde{P}_3(k_1, k_2, t) \\ \tilde{P}_4(k_1, k_2, t) \\ \tilde{P}_5(k_1, k_2, t) \end{pmatrix} = \begin{pmatrix} c_1 & \gamma/2 & 0 & 0 & 0 & \gamma/2 \\ \gamma/2 & c_2 & \gamma/2 & 0 & 0 & 0 \\ 0 & \gamma/2 & c_3 & \gamma/2 & 0 & 0 \\ 0 & 0 & \gamma/2 & c_1^* & \gamma/2 & 0 \\ 0 & 0 & 0 & \gamma/2 & c_2^* & \gamma/2 \\ \gamma/2 & 0 & 0 & 0 & \gamma/2 & c_3^* \end{pmatrix} \begin{pmatrix} \tilde{P}_0(k_1, k_2, t) \\ \tilde{P}_1(k_1, k_2, t) \\ \tilde{P}_2(k_1, k_2, t) \\ \tilde{P}_3(k_1, k_2, t) \\ \tilde{P}_4(k_1, k_2, t) \\ \tilde{P}_5(k_1, k_2, t) \end{pmatrix}, \quad (\text{B8})$$

where

$$c_1 = \frac{1}{3} [\cos(2ak_1) + \cos(ak_1 + bk_2) + \cos(ak_1 - bk_2)] - 1 - \gamma + i2\epsilon \sin(2ak_1), \quad (\text{B9})$$

$$c_2 = \frac{1}{3} [\cos(2ak_1) + \cos(ak_1 + bk_2) + \cos(ak_1 - bk_2)] - 1 - \gamma + i2\epsilon \sin(ak_1 + bk_2), \quad (\text{B10})$$

$$c_3 = \frac{1}{3} [\cos(2ak_1) + \cos(ak_1 + bk_2) + \cos(ak_1 - bk_2)] - 1 - \gamma + i2\epsilon \sin(ak_1 - bk_2). \quad (\text{B11})$$

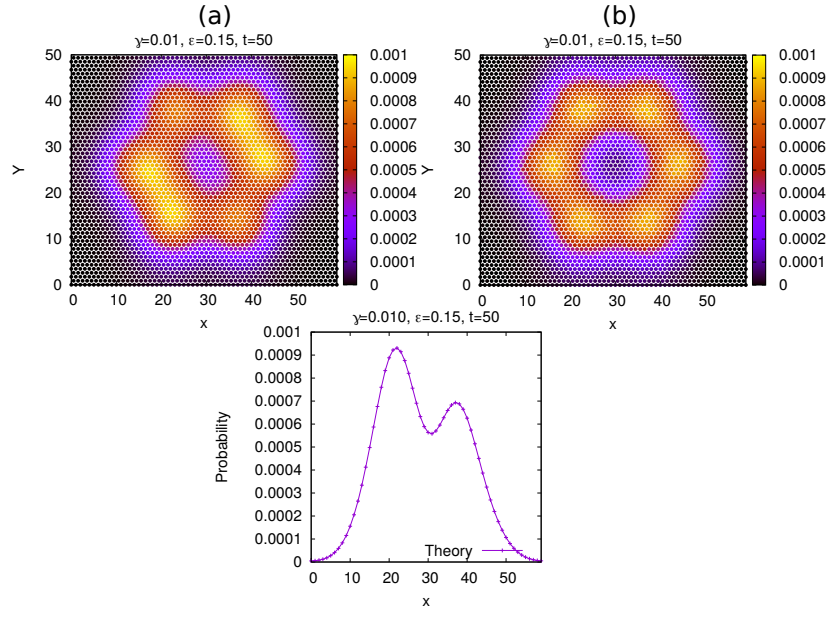


FIG. 11. Comparison of (a) theory and (b) kinetic Monte Carlo result of active random walk on triangular lattice. [NOT MATCHING]